

Interface Control Document

STX-01-00017

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1 Document Control

Revision	Date	Section	Description
A	2016-01-20	All	First Release
B	2016-04-01	8	Second Release
C	2020-04-02	All	Third Release

2 Revision Control

Product	Part Number	Revisions Covered	Notes
STX	STX-01-00017	C	

3 Related Documents

No.	Document Name	Document Reference
OPT-01-00017	Options Sheet: STX	Rev C
USM-01-00017	User Manual: STX	Rev C

4 Nomenclature/Definitions

4.1 Abbreviations

ADC	Analogue-to-Digital Converter
CSK	CubeSat Kit
FIFO	First In First Out
FPGA	Field Programmable Gate Array
OBC	On Board Computer
PA	Power Amplifier
SPI	Serial Peripheral Interface
STX	S-Band Transmitter
SMPS	Switched-mode power supply
Tx	Transmit

5 Introduction

The STX is an integrated RF data transmitter module operating in the S-band and supporting data rates of up to 2 Mbps. This document describes the interfaces between the transmitter and an OBC via the PC/104 header stack connector. A general overview of the intended operation for the transmitter is provided.

6 Overview

An overview of the transmitter is illustrated in Figure 1. It is a compact S-band Transmitter designed for CubeSat nanosatellite missions. It is compatible with the CubeSat nanosatellite standard, with a CubeSat Kit PC/104 form factor. The transmitter implements QPSK or OQPSK modulation with Intelsat IESS-308 based encoding which ensures compatibility with low cost amateur ground segment receivers. The transmitter's frequency of operation is selectable from 2.40 - 2.45 GHz (amateur band). The frequency of operation is user selectable within the band. The carrier frequency is adjustable in 500 kHz steps. Output power is adjustable in 2 dB steps from 24 dBm to 30 dBm. The transmitter is configured via an I²C data bus and high speed payload data is sent via SPI. Data transmission rates of up to 2 Mbps are supported with $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ rate modes. The transmitter is powered from the battery bus or 5V, and can be powered down externally via an enable line from the PC/104 header.

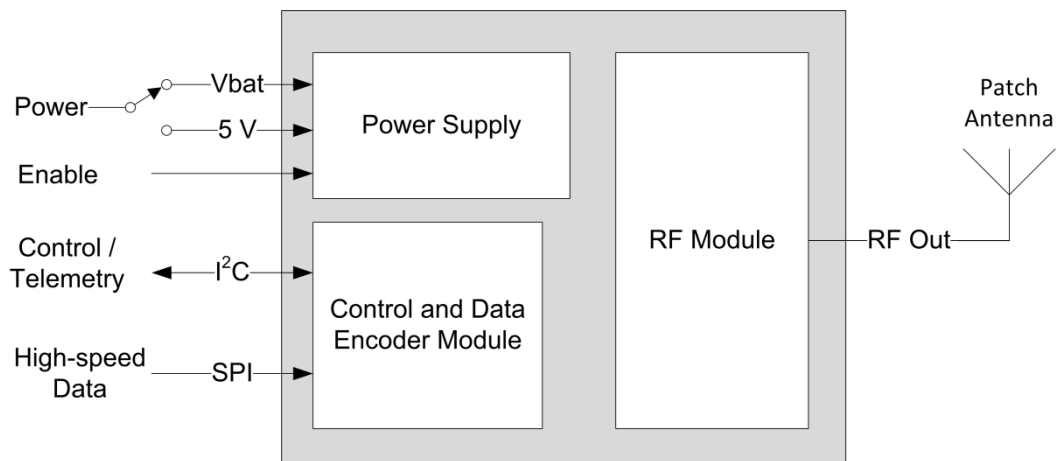


Figure 1: System block diagram.

6.1 Encoding and modulation overview

- QPSK and OQPSK modulation schemes.
- Open Network Encoding scheme based on IntelSAT IESS-308 specifications.
- V.35 IntelSAT scrambler.
- Differential encoding.
- Half-rate convolutional encoding ($K=7$).
- Pulse shaping filter (0.35 roll-off factor).
- 2 Mbps (2,000,000 bps) data rate with full, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ data rate modes.
 - 1 Mbps maximum user data as half-rate convolutional encoding is implemented.

A diagram detailing the flow of data is provided in Figure 2. A detailed description of the V.35 scrambler, differential and $\frac{1}{2}$ rate convolutional encoder implementation is provided in the Appendix.

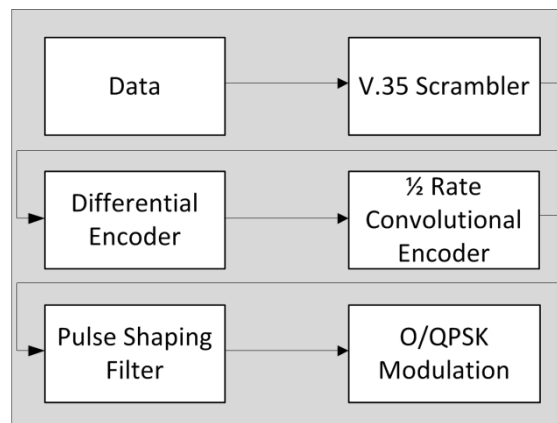


Figure 2: Data flow diagram.

7 Absolute maximum ratings

Parameter	Notes	Value	Unit
Supply Voltage(s)	BatV option only. Regulated from Battery bus.	14	V
	5V option only. From a regulated 5V supply.	5	V
STX_EN	STX Enable	-0.3 to 5	V
Operating temperature		-25 to +61	°C
Storage Temperature		-40 to +85	°C

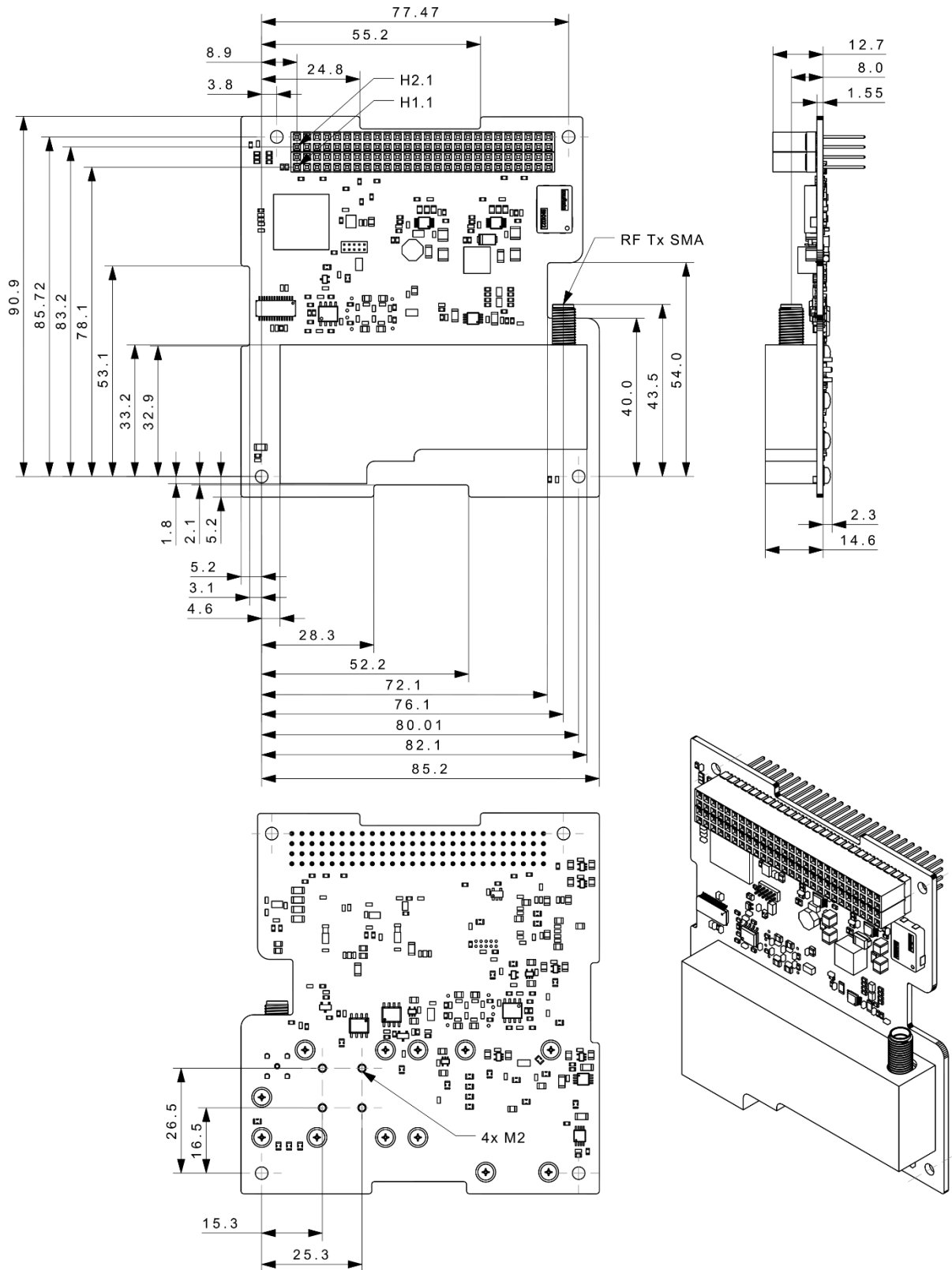
8 Electrical characteristics

Parameter	Notes	Min	Typ	Max	Unit
Power					
BatV (7.2 V)					
Voltage(s)	From a regulated battery bus	6	7.2	12	V
Current	STX_EN = 1	-	84	-	mA
Current	STX_EN = 0	-	0	-	mA
5 V					
Voltage(s)	From 5 V bus	-	5	-	V
Current	STX_EN = 1	-	104	-	mA
Current	STX_EN = 0	-	0	-	mA
PA Enabled		5V		BatV	
Current (Typical)	24 dBm RF out	0.50		0.37	A
	26 dBm RF out	0.57		0.43	A
	28 dBm RF out	0.68		0.52	A
	30 dBm RF out	0.84		0.63	A
DC Power					
Idle power	Transmitter OFF	-	610	-	mW
Transmit 24 dBm RF	Transmitter ON	-	2.7	-	W
Transmit 26 dBm RF	Transmitter ON	-	3.1	-	W
Transmit 28 dBm RF	Transmitter ON	-	3.7	-	W
Transmit 30 dBm RF	Transmitter ON	-	4.6	5	W
RF characteristics					
Transmitter					
Frequency range	Amateur band	2.40	-	2.45	GHz
Output power	2 dB increments	24	-	30	dBm
Frequency stability		-	±2.5	-	ppm
Channel spacing		-	500	-	kHz
SNR		-	>20	-	dB

I²C					
SCL frequency		50	400	500	kHz
Node address		-	0x26	-	hex
Address scheme		-	7	-	bit
SPI					
Transmission speed		-	4	8	Mbps
PC/104 I/O characteristics (excluding STX_EN, nRESET)					
V _{IL} – Low-level input voltage		-0.3	-	0.8	V
V _{IH} – High-level input voltage		2	-	3.6	V
V _{OL} – Low-level output voltage		-	-	0.4	V
V _{OH} – High-level output voltage		2.4	-	-	V
PC/104 I/O characteristics (STX_EN)					
V _{IL} – Low-level input voltage		-	-	0.4	V
V _{IH} – High-level input voltage	Includes a pull-up resistor	1.8	-	3.3	V
PC/104 I/O characteristics (nRESET)					
V _{IL} – Low-level input voltage		-	-	0.9	V
V _{IH} – High-level input voltage	Includes a pull-up resistor	2.1	-	3.3	V

9 Mechanical characteristics

Parameter	Notes	Min	Typ	Max	Unit
Physical					
Dimensions	See diagrams				
Weight		--	<100	--	g
Output ports					
RF connector	Tx = SMA				
Torque specifications					
SMA connector	SMA torque wrench (5/16")		90 (8)		N·cm (in-lb)



10 Materials List

Material	Manufacturer	%TML	%CVCM	%WVR	Application	Notes
Scotch-Weld 2216 Epoxy B/A	3M	0.77	0.04	0.23	Adhesive Fixing	-
PCB material	FR4	0.62	0	0.1	-	Note: worst case on NASA out-gassing list
Solder Resist	CARAPACE EMP110	0.95	0.02	0.31	-	-
Solder	Sn62 or Sn63	-	-	-	-	-
Flux	NASCO NCF	-	-	-	-	Non-corrosive
Aluminium 6082	-	-	-	-	Enclosure	Conversion coated to MIL-C-5541 Type 1 Class 3

11 Interfaces

11.1 Mechanical interfaces

11.1.1 Heat sink interface

Provision is made for a heat sink interface on the underside of the PCB. The RF enclosure provides four threaded pillars as depicted on the mechanical diagram which allow for M2 screws to fasten a heat sink to the unit. The thread in the pillars is approximately 4 mm deep. A simple L-shaped heat sink that attaches to the inside of the satellite structure is suggested. CHO-THERM® may be used between the board and the heat sink for improved thermal conductivity.

11.2 Hardware interfaces

11.2.1 CSK header connections

Figure 4 illustrates the connections that are made available at the CSK header. Broken lines indicate optional connections. The only non-optional connection to the header (excluding power signals) is I²C and SPI. Optional connections that are provided on the header include a transmit ready (TR) signal, FPGA reset and STX enable. The optional connections can be selected at the time of production and should be selected according to application and performance requirements. Should the optional functionality not be required it will not be made available at the header (there will be no physical connection). An explanation of the various connections is detailed later within this document. Refer to Section 8 for electrical characteristics of the various signals.



Important to note is that the grouped power and ground pins (i.e. those consisting of more than one pin) are shorted at the PC/104 connector regardless of whether an alternate pin is selected. Care should be taken when utilising these pins for other signals on the bus as this may cause a short circuit condition if more than one signal is used on any of the grouped pins.

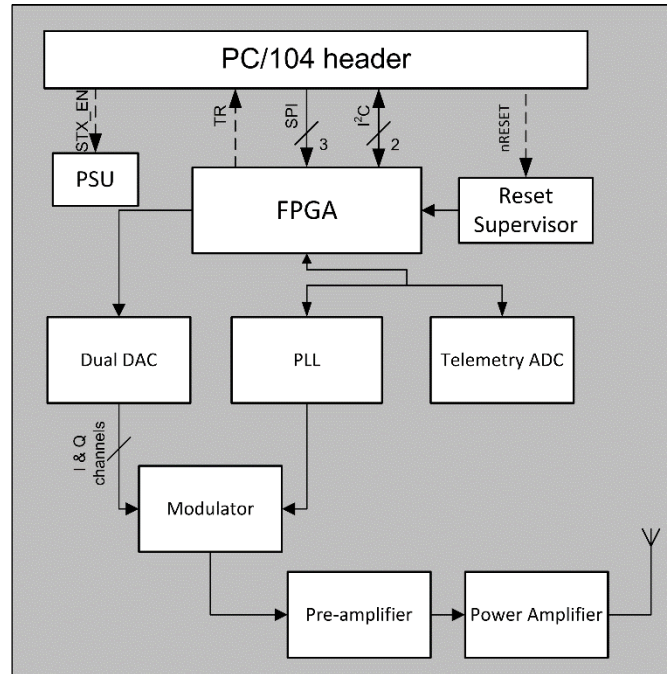


Figure 4: Block diagram of the CSK header connections. Broken lines indicate optional connections.

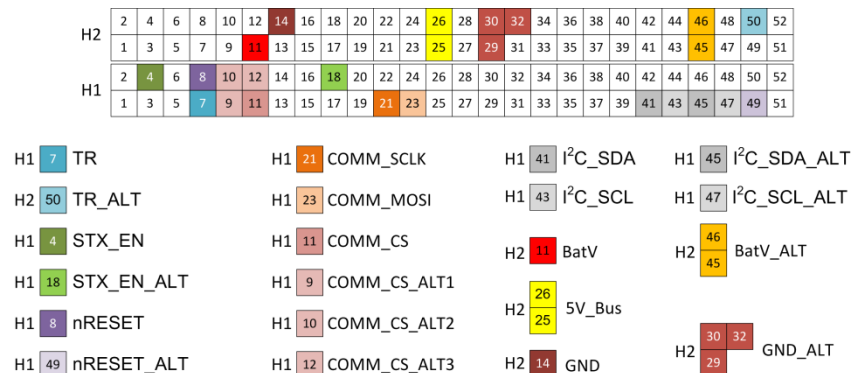


Figure 5: PC/104 header pinouts.

Table 1: CSK connector pinouts

Signal name	Primary pin(s)	Alternate pin(s)	I/O type	Description	Optional
SDA	H1.41	H1.45	Bidirectional	I ² C serial data	No
SCL	H1.43	H1.47	Input	I ² C serial clock	No
COMM_SCLK	H1.21		Input	SPI Clock	No
COMM_MOSI	H1.23		Input	SPI Data	No
COMM_CS	H1.11	H1.9 or H1.10 or H1.12	Input	SPI Chip Select	No
nRESET	H1.8	H1.49	Input	FPGA reset (active low)	Yes
TR	H1.7	H2.50	Output	Transmit ready	Yes
STX_EN	H1.4	H1.18	Input	Board power enable	Yes
BatV ¹	H2.11	H2.45, H2.46	Power	Battery bus supply	Yes
5V_BUS ¹	H2.25, H2.26		Power	Optional 5 V supply	Yes
GND ¹	H2.14	H2.29, H2.30, H2.32	Power	Power ground	No

NOTE 1: Grouped power and ground pins, regardless of whether the primary or alternate is selected for manufacture, remain shorted at the PC/104 connector. Care should be taken when using these pins for other signals on the bus.

11.2.2 FPGA reset

A reset signal is provided to the header that will allow an external subsystem such as an OBC to reset the FPGA to a known good state. This is an optional signal. Alternatively cycling the power of the radio or using the soft reset register will also place the FPGA into a known good state.

11.2.3 RF connector

A 50 Ω SMA connector is used as the RF output port of the transmitter. An antenna or RF load with return loss of -10 dB or better is required to ensure the power amplifier has an adequate terminating load to prevent damage. When not transmitting into an antenna, ensure that an appropriate high-power RF load is connected to prevent damaging the power amplifier. The SMA connector can be torqued to 8 in-lb (90 N·cm).

11.2.4 Coaxial interfaces

There seems to be inconsistent usage of the terms plug and jack. For the purpose of this document, a plug will refer to connectors that have a moving nut that fits over a connector that has no moving parts (jack).

The transmitter's output coaxial interface is an angled SMA board mounted female SMA similar to Telegartner part number J01151A0191. The coaxial cable harness that will interface with the transmitter connector must terminate in an SMA cable crimped plug. A right angle plug is advised, such as Telegartner part number J01150A0061 as a straight crimp connector will not allow enough space for the cable to bend before the CubeSat Kit header connector.

11.2.5 STX enable

The STX enable line allows the board to be powered down when not in use. This is a hard line connected to the PC/104 header and its functionality is optional and needs to be configured during manufacturing. The transmitter may be configured to always be enabled if so desired, negating the use of the enable line. The enable line includes a pull-up; pull low to disable the transmitter. Float to enable or apply voltage referred to in Section 8 for STX_EN to enable the board.

11.3 Software interfaces

11.3.1 I²C operation

All data, telecommands and telemetry are communicated via I²C. The default I²C address is 0x26, but can be configured according to user specifications at time of production (Refer to Options Sheet for default). Issuing a telecommand (writing data) has the following procedure. The first byte of an I²C transaction written points to the address of the register, with the following byte(s) writing the value to the register. Reading telemetry follows a similar approach, firstly a byte is written to point to the correct register followed by a read transaction to return the value. Consecutive read transactions automatically increment the read pointer.

11.3.2 Modes of operation

The transmitter has three basic modes of operation:

- Configuration mode
- Synchronisation mode
- Data mode
- Test data mode

11.3.2.1 Configuration mode

At power-up, the transmitter is in configuration mode. The I²C interface should be used to send telecommands to the transmitter to set it up prior to the transmission of data (data rate, transmit frequency, transmit power, etc).

11.3.2.2 Synchronisation mode

After configuration, a telecommand may be sent to put the transmitter into Synchronisation Mode. In this mode, when the PA is activated, synchronisation bytes will be sent from the transmitter (no real data is read from the SPI input data FIFO). This allows the ground station receiver to achieve lock (synchronisation) before actual payload data is transmitted over the link. In this mode, the data buffer will accept data via the SPI interface only until the internal FIFO is full. The synchronisation word is a CCSDS 32-bit Attached Sync Marker (ASM) for non-turbo-coded data: 0x1ACFFC1D.

11.3.2.3 Data mode

In Data mode, data from the SPI input FIFO is transmitted. The transmitter should only be placed into Data mode when the SPI FIFO contains data (indicated by the transmit ready line low) - this will void an initial buffer underrun condition. Each SPI bus transaction transfers one (or more) byte(s) of data (a maximum of 4096 bytes may be transferred per SPI transaction). The buffer is 4096 bytes deep. The source of the SPI data (On-Board Computer or Mass Storage) should monitor the Transmit

ready (TR) line, and send data as quickly as possible if the TR line is high, to ensure that a buffer underrun does not occur (FIFO empty condition). The TR line is active (high) when there are less than 513 bytes in the buffer. It is recommended that the SPI data source sends data to the transmitter at a faster rate than the user data rate. The user data rate is half the over-the-air data rate due to the $\frac{1}{2}$ rate convolutional encoding. If a buffer underrun occurs, *dummy data* (0x55) will be transmitted.

11.3.2.4 Test data mode

Test data mode will continuously encode and send a test counter of data. The counter begins at 0x00 and continues until 0xFF, before beginning again. In this mode, the data buffer will accept data via the SPI interface only until the internal FIFO is full.

11.3.3 Recommended procedure for transmitting data

A recommended procedure for transmitting user data with the transmitter is as follows:

- Ensure that a 50 Ω load is connected at all times.
- Configure transmitter registers (Power level, data rate...).
- Place transmitter into Synchronisation mode with PA on.
- Fill buffer with 4096 bytes of user data.
- Allow adequate time for Synchronisation to occur with ground station.
- Place transmitter into Data mode.
- Monitor the transmit ready line; transfer additional data to the buffer as required.
- Request telemetry relating to overruns, underruns and the buffer count as a source of transmit verification.
- Place transmitter into Configuration mode with the PA off.

Notes:

In order to free up the OBC for the maximum amount of time when the transmitter is in Data mode, as soon as the transmit ready flag indicates that there are 512 bytes or less of data in the buffer, it should be topped up to as close to 4096 bytes as possible all the while considering the buffer is being decreased at the rate of the user data rate. Care should be taken to not incur any overruns; optimal timing can be established by monitoring the buffer count and buffer overrun registers.

On entering Data mode from Synchronisation, Test Data or Configuration mode, a filler byte (0x55) is first transmitted, indicating a change in the mode, which is then followed by the user data.

When the buffer is filled with 4096 bytes of data, it typically takes 33 ms to transmit all the data from the buffer, at full data rate.

If the transmitter is in Data mode with the PA off, the buffer will still send data out and operate as per usual, yet the data will not actually be transmitted to the ground station as the PA is not active.

The transmitter has a 15 minute failsafe timeout which disables the PA if it is left on accidentally. The timer operates independently from the other control logic and as such the I²C Control register will indicate that the PA is on. The PA would need to be disabled first before trying to use it again once a timeout has occurred. When in Configuration mode or entering Configuration mode, the timer is held in reset.

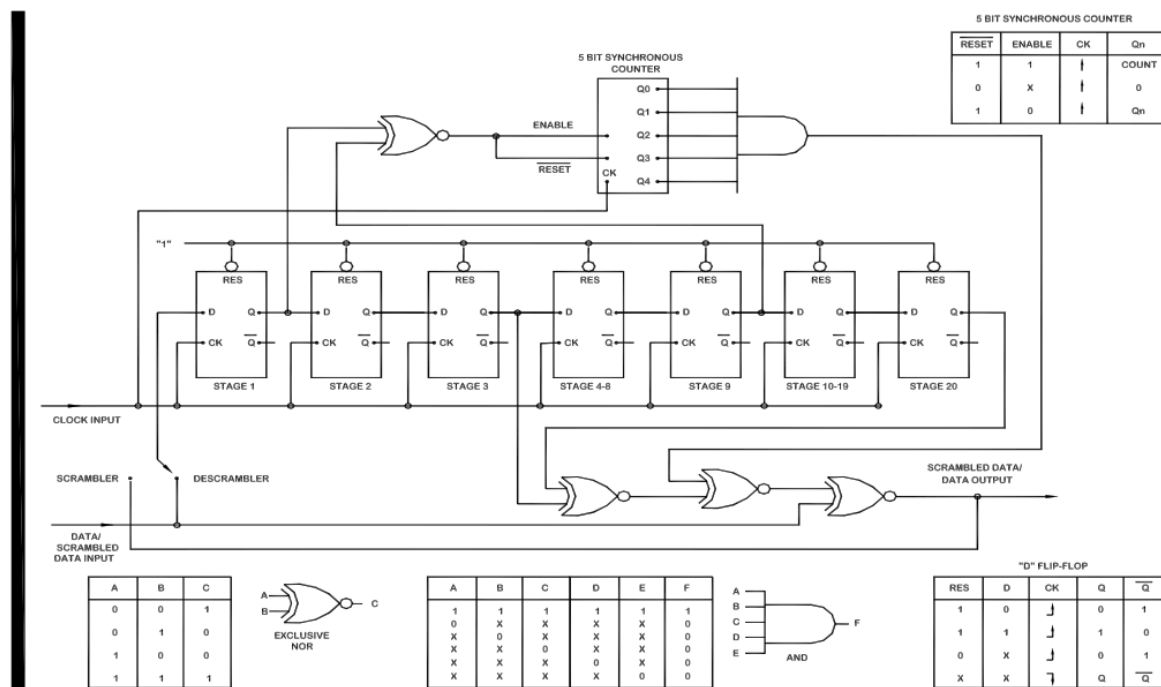
A Appendix

The following figures were sourced from the specification document INTELSAT EARTH STATION STANDARDS (IESS) Document IESS-308 (Rev. 10), and details the modules that have been implemented.

A.1 V.35 Scrambler

IESS-308 (Rev. 10)
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Figure 24
Scrambler/Descrambler Logic Diagram



P:\CAD\IESS\308NEW\308-24.DWG
TITLE: SCRAMBLER/DESCRAMBLER LOGIC DIAGRAM

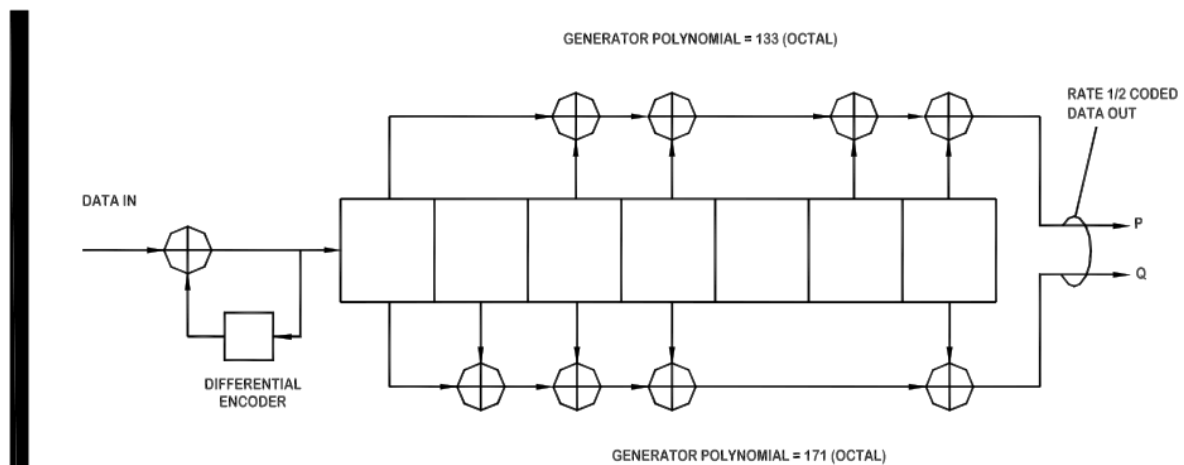
Figure 6: V.35 Scrambler logic diagram. Source: IESS-308 (Rev. 10) document.

A.2 Differential and half rate convolutional encoder

IESS-308 (Rev. 10)

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Figure 10
Convolutional Encoding Process Block Diagrams
For Use With Viterbi Decoding
(Rate 1/2 and 3/4 FEC)



NOTES:

1. THE SYMBOL $\oplus$ DENOTES A MODULO-2 ADDER.
2. IN THE DELETING BIT PATTERN, 1 DENOTES TRANSMITTING AND 0 DENOTES DELETING.
3. THE RIGHT-MOST SHIFT REGISTER STAGE CORRESPONDS TO THE LEAST SIGNIFICANT BIT IN THE GENERATOR POLYNOMIAL.

Figure 7: Differential and 1/2 rate convolutional encoder. Source: IESS-308 (Rev. 10) document.